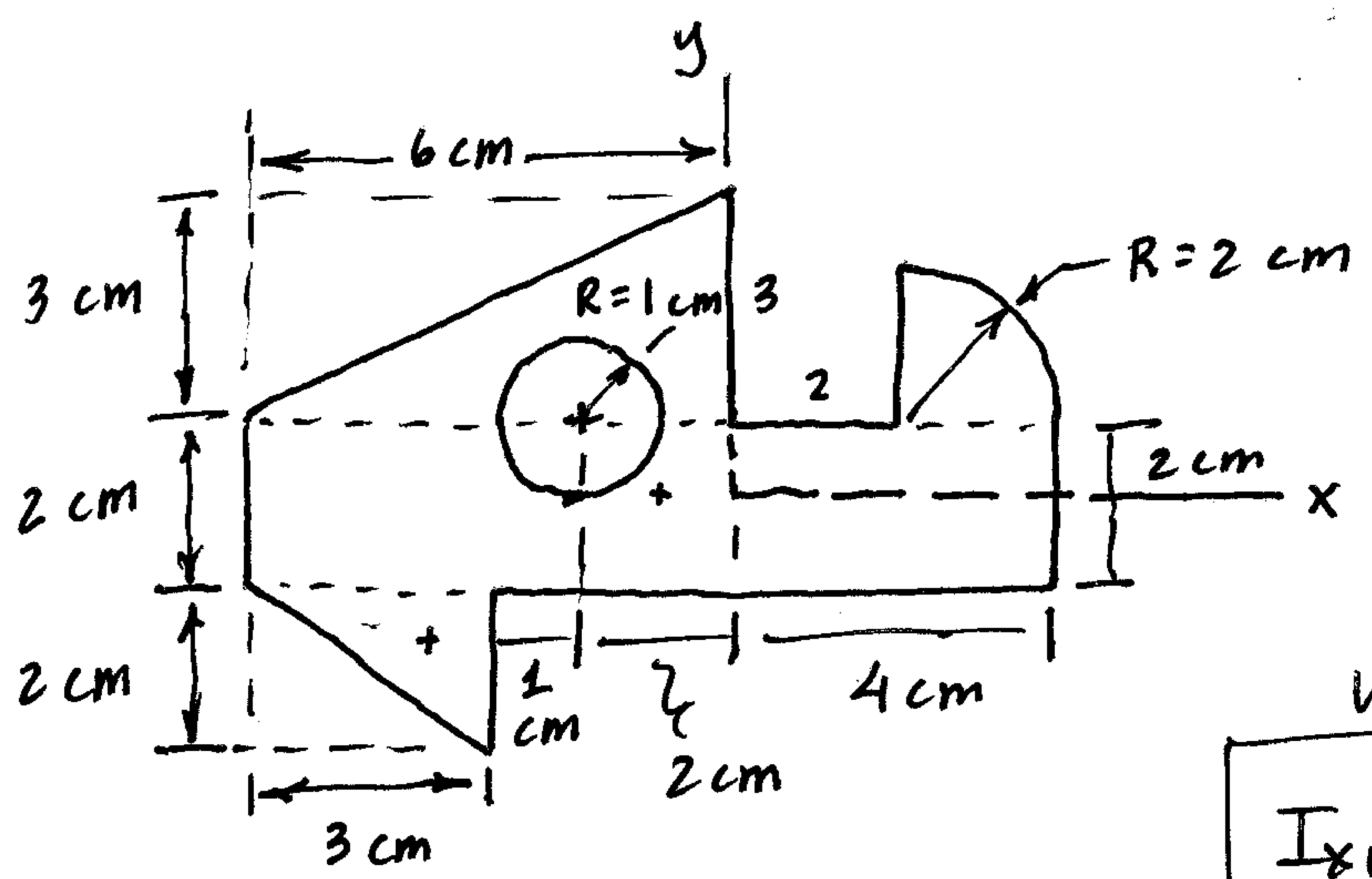
POI
Ex Prob 5

Determine I_{xy} for this shape



We previously determined:

$$\boxed{I_x = 63.9 \text{ cm}^4}$$

$$\boxed{I_y = 303.2 \text{ cm}^4}$$

use PAT:

$$\boxed{I_{xy} = \Sigma(\bar{I}_{x'y'} + A\,dx\,dy)}$$

R = 2

$\frac{4R}{3\pi} = \frac{8}{3\pi}$

$= .8488$ cm

This problem was specifically chosen because it illustrates how

- $\bar{I}_{x'y'}$ can be 0, −, or +
- dx and dy can be + or −
- Holes are subtracted, i.e. $-[\bar{I}_{xy} + A\,dx\,dy]$

$$I_{xy} = \Big[0 + \underbrace{(10)}_{A}\underbrace{(2)}_{dx}\underbrace{(-1)}_{dy}(0)\Big]$$

$$+ \Big[\underset{\text{Positive!}}{+}\tfrac{1}{72}(6)^2(3)^2 + \underbrace{\tfrac{1}{2}(6)(3)}_{A}\underbrace{(-2)}_{dx}\underbrace{(+2)}_{dy}\Big]$$

Neg!

$$+ \Big[-\tfrac{1}{72}(3)^2(2)^2 + \tfrac{1}{2}(3)(2)\underbrace{(-1.67)}_{d_y = -(1 + 2/3)}\underbrace{(-4)}_{dx}\Big]$$

$$+ \Big[-.01647(2)^4 + \frac{\pi(2)^2}{4}(\overset{+}{2}.8488)(\overset{+}{1}.8488)\Big]$$

Neg!

Hole!

$$-\Big[\underset{I=0}{0} + \pi(1)^2\underbrace{(-2)}_{dx}\underbrace{(1)}_{dy}\Big]$$

$I_{xy} = 0$

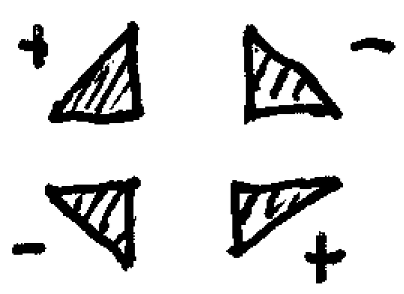
$-31.5 + 19.5$

$+ 16.28$

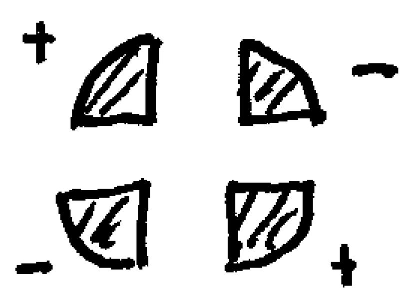
$+ 6.28$

$$\boxed{I_{xy} = 10.56 \text{ cm}^4}$$

Remember:

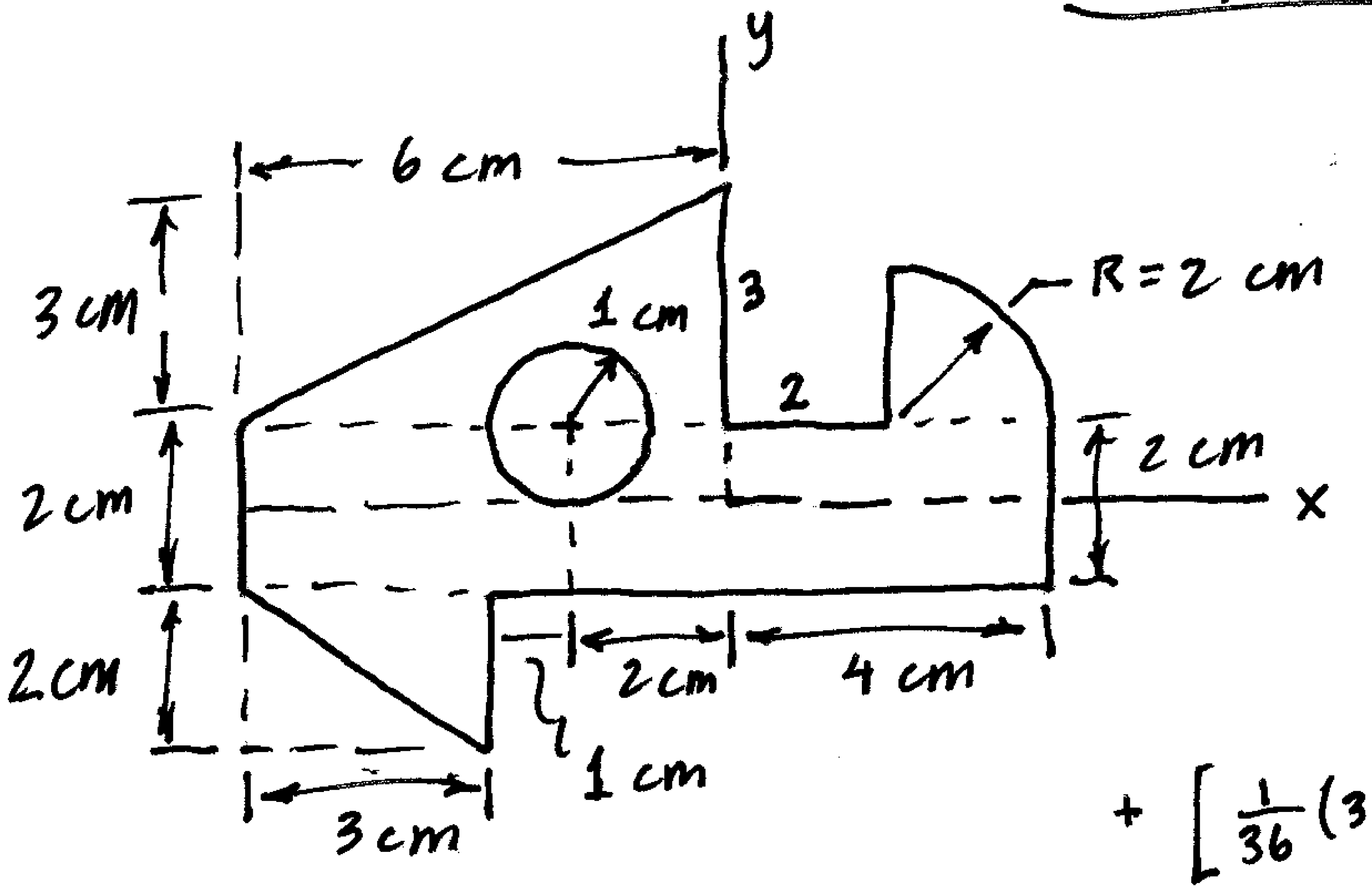


$$\boxed{\bar{I}_{x'y'} = (-1)^n \tfrac{1}{72} b^2 h^2}$$



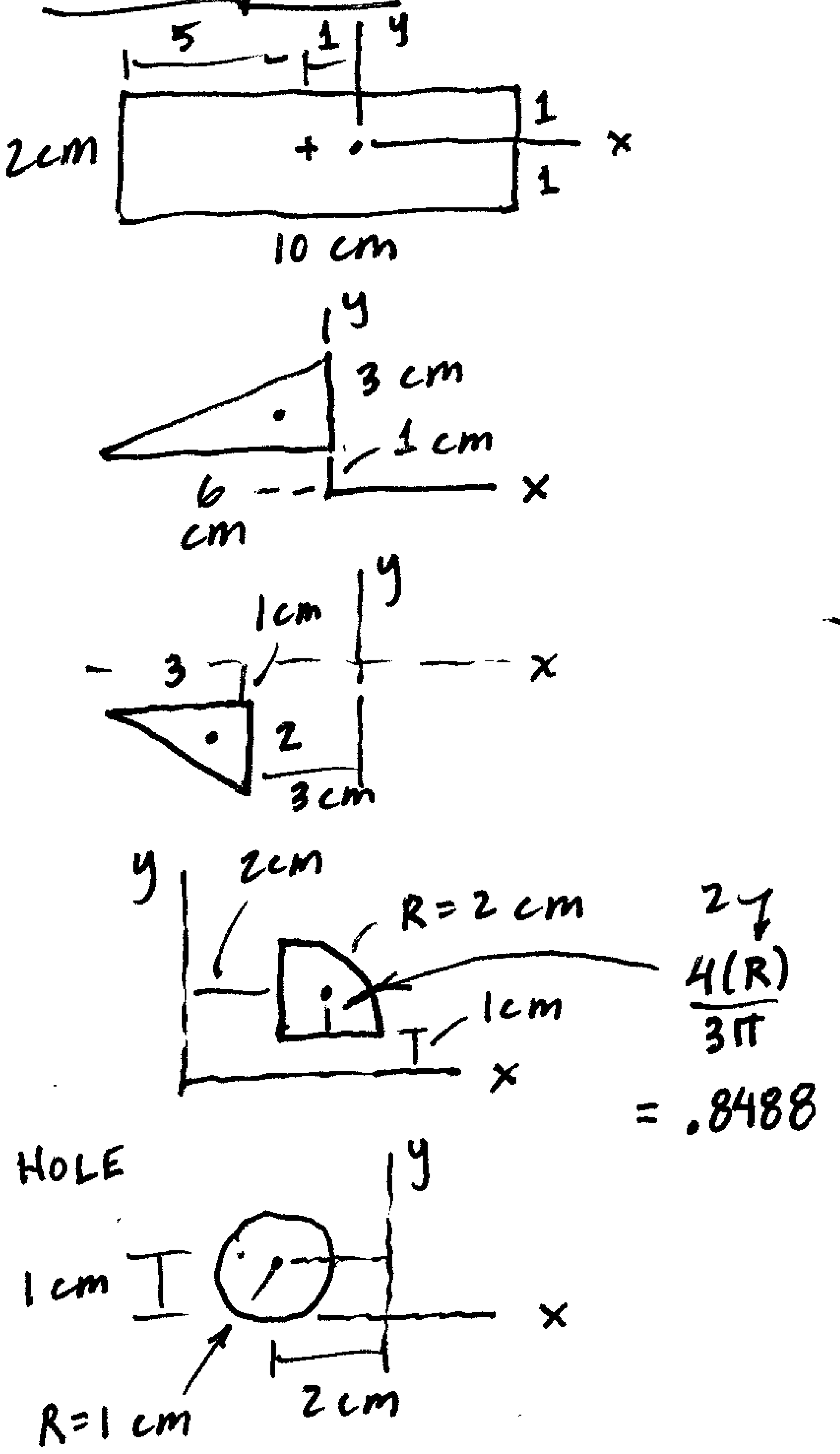
$$\boxed{\bar{I}_{x'y'} = (-1)^n .01647 R^4}$$

Composite MOI, POI Problem 5



Determine: I_x, I_y, I_{xy}

Shapes



$$I_x = \left[\tfrac{1}{12}(10)(2)^3\right]_{\square} + \left[\tfrac{1}{36}(6)(3)^3 + \tfrac{1}{2}(6)(3)(1+1)^2\right]_{\triangle} + \left[\tfrac{1}{36}(3)(2)^3 + \tfrac{1}{2}(3)(2)(1+2/3)^2\right]_{\triangle} + \left[.05488(2)^4 + \frac{\pi(2)^2}{4}(1+.8488)^2\right] - \left[\frac{\pi}{4}(1)^4 + \pi(1)^2(1)^2\right]_{\circ}$$

$$I_x = 6.667 + 40.5 + 9.0 + 11.62 - 3.927$$

$$\boxed{I_x = 63.9\ cm^4}$$

$$I_y = \left[\tfrac{1}{12}(2)(10)^3 + 10(2)(1)^2\right]_{\square} + \left[\tfrac{1}{12}(3)(6)^3\right]_{\triangle} \leftarrow I_{base}\ ! + \left[\tfrac{1}{36}(2)(3)^3 + \tfrac{1}{2}(3)(2)(3+1)^2\right]_{\triangle} + \left[.05488(2)^4 + \frac{\pi(2)^2}{4}(2+.8488)^2\right] - \left[\frac{\pi}{4}(1)^4 + \pi(1)^2(2)^2\right]_{\circ}$$

$$I_y = 186.67 + 54 + 49.5 + 26.37 - 13.35$$

$$\boxed{I_y = 303.2\ cm^4}$$

$$I_{xy} = \Sigma\left[\bar{I}_{xy} + A\,d_x d_y\right] = \left[0 + 10(2)(-1)(0)\right]_{\square} + \left[+\tfrac{1}{72}(6)^2(3)^2 + \tfrac{1}{2}(6)(3)(-2)(+2)\right]_{\triangle} + \left[-\tfrac{1}{72}(3)^2(2)^2 + \tfrac{1}{2}(3)(2)(-1.67)(\overbrace{\ }^{-(3+1)})\right]_{\triangle} + \left[-.01647(2)^4 + \frac{\pi(2)^2}{4}(2.8488)(1.8488)\right] - \left[0 + \pi(1)^2(-2)(1)\right]_{\circ}$$

$$I_{xy} = 0 - 31.5 + 19.5 + 16.28 + 6.28 \Rightarrow \boxed{I_{xy} = 10.56\ cm^4}$$